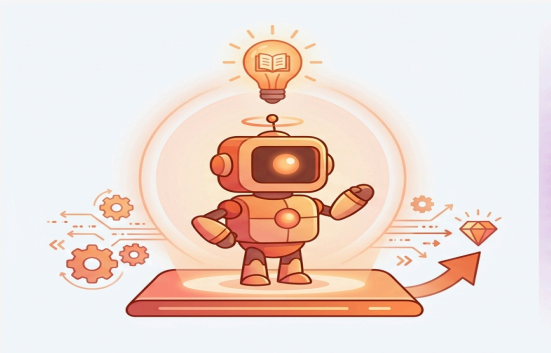


More Agents, Less Accuracy: When Multi-Agent LLM Voting Hurts on Knowledge-Bound Tasks

COMMON PRACTICE

"More Agents Is All You Need" — Li et al., 2024



↑ more accuracy (claimed)

Monotone gains reported on broad benchmarks ⇒ default architectural recipe

BUT

OUR FINDING

On knowledge-bound NLP, voting hurts accuracy

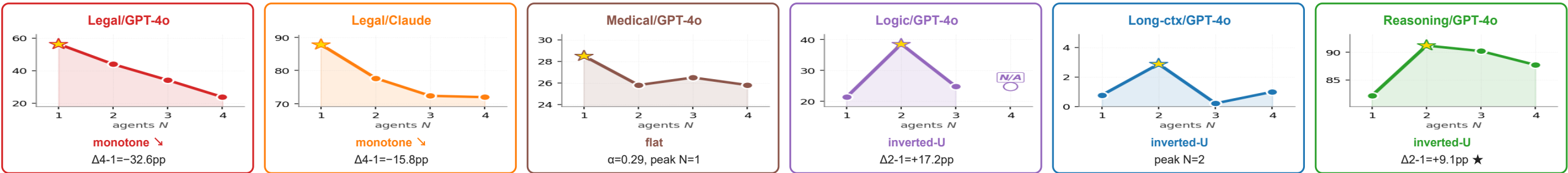


↓ LESS accuracy

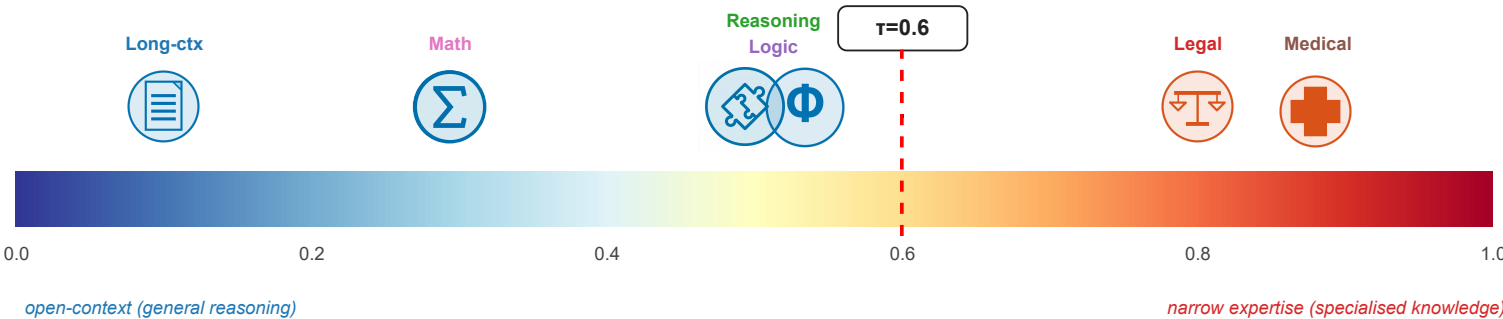
-32.6 pp

on legal/GPT-4o (N=1→4, McNemar $p < 10^{-100}$)

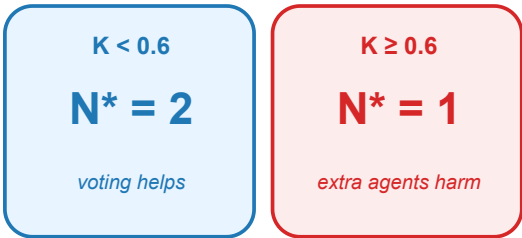
Evidence: same protocol, different shapes — six (domain × model) cells



Recipe: predict N^* from a single text-judged feature K in ≈ 15 min



Decision rule



Verify with paired pilot: $r > \alpha(1-p) / (1-\alpha)$

$$\Delta = (1-\alpha) \cdot r - \alpha \cdot (1-p)$$

7/8 main cells (LOO); 9/12 combined w/ tier-2

K scored from task description by two LLM judges (inter-judge $r=0.885$).